

tmux documentation

basic-digital tmux — CLAUDE.md

INHERITED FROM constitution/CLAUDE.md

All rules in `constitution/CLAUDE.md` and the `constitution/Constitution.md` it references apply **unconditionally** to this project. Project-specific rules below **extend** them — they do NOT weaken or override any universal clause. When this file disagrees with the constitution submodule, **the constitution wins**.

@constitution/CLAUDE.md

MANDATORY ANTI-BLUFF END-USER-QUALITY COVENANT

Forensic anchor — verbatim user mandate (2026-04-28, reasserted 2026-05-21):

"We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

Operative rule. The bar for shipping is **not** "tests pass" but **"end users can actually use the feature."** Every PASS — test OR HelixQA Challenge — MUST carry positive captured runtime evidence that the feature works for the end user. Metadata-only PASS, configuration-only PASS, "absence-of-error" PASS, and grep-without-runtime PASS are all critical defects regardless of how green the summary line looks. Tests and Challenges are bound EQUALLY.

This covenant is restated verbatim in every governance file at the consumer layer (`CLAUDE.md`, `AGENTS.md`, `QWEN.md`) so any tool that does not expand `@imports` still reads it. The same verbatim block lives upstream at `constitution/Constitution.md` §11.4, `constitution/CLAUDE.md`, and `constitution/AGENTS.md`. Removing or weakening this block is a §11.4-class release blocker.

Canonical authority: `constitution/Constitution.md` §11.4 and its sub-anchors §11.4.1 through §11.4.78. Project anchor: [Constitution.md](#) §101.

Non-compliance is a release blocker regardless of context.

Read first on a fresh conversation: `CONTINUATION.md` (§0/§3/§8 are mandatory), then `Issues.md` for OPEN/PARTIAL/BLOCKED/RUNNING items. Canonical project authority: [Constitution.md](#) (Project Articles §101–§109) which extends `constitution/Constitution.md`.

Project overview

Verified hardened tmux 3.6a build with jemalloc, OOM protection, and per-session isolation via the tmx wrapper. Built around a hard verification gate that refuses to expose the binary unless functional tests pass. **Native dual-OS** (since 2026-05-13): runs as a host process on Linux AND macOS — no VM in the daily-use path. Each `tmx new -s NAME` creates its own tmux server with OS-native isolation: cgroup-v2 transient scope (`systemd-run --user --scope`) on Linux; POSIX rlimit wrapper (`RLIMIT_CPU + RLIMIT_NPROC`) on macOS. The session shell is the operator's host shell with full `$PATH`. Honest gap: `RLIMIT_AS` (memory) is NOT enforced by XNU for unprivileged processes — see `docs/guide/README.md §5.6`.

Critical base rules restated (for sessions that don't expand @imports)

- **No bluffing.** Every PASS — test OR HelixQA Challenge — carries positive runtime evidence (cgroup//proc file *content*, capture-pane output, kernel log lines), never just an exit code. Universal §11.4 / Project §101.
- **FAIL-bluffs equally forbidden.** A test that exits FAIL for a script bug (not a product defect) is fixed at the source layer, never at call sites. Universal §11.4.1.
- **No-guessing.** Never use likely/probably/maybe/seems/appears in cause descriptions — prove with forensic evidence or mark UNCONFIRMED: / PENDING_FORENSICS:. Universal §11.4.6.
- **Operator-path coverage.** Every gate test exercises the SAME entry point an end-user invokes (`tmx new -s X`, not hand-spawned equivalents). Project §102.
- **Four-layer coverage.** Every defect lands all four layers before the cycle closes. Project §103.
- **CONTINUATION.md** updated in the SAME commit as any non-trivial state change. Universal §12.10.
- **commit_all.sh only.** Never `git push` directly on the main repo.

Commands (exact)

Step	Command
Install build deps (one-time; Linux needs root, macOS uses brew)	<code>bash scripts/install_deps.sh</code>
Full pipeline	<code>bash scripts/setup.sh</code>
Build only	<code>bash scripts/setup.sh --build-only</code>
Verify only	<code>bash scripts/setup.sh --verify-only</code>
Verification gate	<code>bash scripts/verify.sh</code>
Run all tests	<code>bash scripts/tests/run_all.sh</code>
Single test	<code>bash scripts/tests/NN_*.sh</code> (zero-padded, 01..17)
Constitution inheritance gate	<code>bash scripts/tests/ test_constitution_inheritance.sh</code>
Meta-test (paired mutation)	<code>bash scripts/tests/ meta_test_false_positive_proof.sh</code>
Containerized test run (bounded subset)	<code>bash scripts/test_containerized.sh</code>
VM test run (full suite)	

Step	Command
	<code>bash scripts/test_vm.sh</code> (<code>TMX_TEST_DESTRUCTIVE=1</code> for tests 12/13/14; <code>META=1</code> for meta-test)
End-to-end automation	<code>bash scripts/test_e2e.sh</code>
Commit + push (github+gitlab)	<code>bash commit_all.sh "message"</code>
Per-session wrapper	<code>tmx {new attach ls kill}</code>

Never `git push` directly — use `commit_all.sh`.

Structure

Path	Role
<code>constitution/</code>	HelixConstitution submodule — universal governance (pinned 7f738df) — do not modify
<code>tmux/</code>	upstream submodule (tag 3.6a) — do not modify
<code>Containers/</code> <code>tmux/build*/</code>	vasic-digital cgroup helpers submodule build output (<code>.gitignore'd</code>)
<code>scripts/</code>	build, verify, install, 17 tests + inheritance gate, challenges, wrapper + conf templates
<code>scripts/tmx</code>	generated dispatcher (<code>.gitignore'd</code>) — from <code>tmx.template</code> (Linux) or <code>tmx-mac.template</code> (macOS bridge)
<code>scripts/tmux.conf.template</code>	the tmux config template (scrollback, copy-mode, Claude-Code TUI passthrough, clipboard)
<code>scripts/tests/</code> <code>meta_test_false_positive_proof.sh</code>	§103 layer-4 paired-mutation harness
<code>scripts/challenges/tmux.yaml</code>	HelixQA Challenge specs
<code>commit_all.sh</code>	only allowed push mechanism
<code>Constitution.md</code>	project authority — Project Articles §101–§109 (extends <code>constitution/</code>)
<code>Issues.md</code>	OPEN / PARTIAL / BLOCKED / RUNNING only
<code>Fixed.md</code>	RESOLVED items with closure SHA + evidence
<code>CONTINUATION.md</code>	live handoff state (must stay fresh)

Project-specific MANDATORY constraints

Build / packaging

- `scripts/setup.sh` detects host OS and invokes the right build pipeline (Linux ELF / macOS Mach-O) — Project §108.

- `scripts/tmux.conf.template` is the source for the generated tmux config; `setup.sh` substitutes OS-specific placeholders (clipboard command). Never hand-edit the generated `~/ .tmux.conf / wrapper`.

Test / verification

- **Test-interrupt-on-discovery** (§103): any defect found during a test cycle STOPS the cycle. Fix at root cause + land all four layers (pre-build gate / runtime test / HelixQA Challenge / paired mutation)
 - rebuild + repeat from the beginning.
- **Operator-path coverage** (§102): gate tests use `tmx new -s X`.
- **Topology dispatch** (§104): tests detect topology and SKIP-with- reason — never silently degrade.
- **Destructive tests** (tests 12/13/14): require `TMX_TEST_DESTRUCTIVE=1` — run only on dedicated test hosts.
- **Constitution inheritance** is itself gated: `scripts/tests/test_constitution_inheritance.sh` + the CM-CONSTITUTION-INHERITANCE paired mutation.

Deployment / release

- `bash commit_all.sh "<msg>"` is the ONLY push mechanism — pushes to github + gitlab.
- `VERSION` is the single source of truth (`version=semver + versionCode=monotonic`). Both bump together with a `CHANGELOG.md` entry on every release.

Project-specific Applied Fixes Reference

#	Fix	Key files
A13	Per-session cgroup isolation (OOM contained per scope) Status bar showed	<code>scripts/tmx.template</code>
A15	<code>claude.exe</code> — strip <code>.exe</code> from window name	<code>scripts/tmux.conf.template</code>
A16	tmux scrolling fix for Claude Code TUI + mobile/Termux	<code>scripts/tmux.conf.template</code> , <code>scripts/tests/17_*.sh</code>
A17	HelixConstitution submodule + verified governance inheritance	<code>constitution/</code> , <code>scripts/tests/test_constitution_inheritance.sh</code>

Full forensic detail: `Fixed.md`.

Files to never edit directly

- `scripts/tmx` — generated by `setup.sh` from `scripts/tmx.template`
 - `tmux/build*/` — build output (in `.gitignore`)
 - `tmux/` — upstream submodule pinned to tag 3.6a
 - `constitution/` — HelixConstitution submodule (governance authority)
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Project overrides of universal rules

None. This project introduces no override of any universal clause.